IoT Based Remote Patient Health Monitoring System

A

Community Project Report

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Submitted in partial fulfilment of the requirements for the Second Year Community Engineering Project

Computer Engineering

Under The Guidance of
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IoT Based Remote Patient Health Monitoring System

During The Academic Year 2025-2026 In The Partial Fulfillment Towards

The Completion of Community Engineering Project in Computer Engineering

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Abstract

In recent years, the integration of Internet of Things (IoT) technology in the healthcare sector has significantly enhanced the efficiency of patient monitoring systems. This project presents the design and implementation of an IoT-based Remote Patient Health Monitoring System, which enables real-time monitoring of vital health parameters such as heart rate, blood oxygen level (SpO_2), and body temperature.

The proposed system utilizes sensors like the MAX30102 for heart rate and SpO_2 measurement, and the DHT11 sensor for temperature sensing, interfaced with an ESP32 microcontroller. The collected physiological data is processed and displayed locally on an LCD screen, while simultaneously being transmitted over Wi-Fi to a cloud-based platform using the Blynk application. This allows healthcare professionals and users to monitor patient health remotely through a web or mobile dashboard.

The system is designed to be cost-effective, portable, and user-friendly, making it suitable for continuous health monitoring, especially in remote or resource-limited areas. It provides timely access to critical health data, thereby enabling early detection of abnormalities and improving patient care.

Overall, this project demonstrates how IoT can be effectively applied in healthcare systems to provide real-time, reliable, and remote monitoring solutions.

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Introduction

1.1 Motivation

The motivation behind developing an IoT-based Remote Patient Health Monitoring System arises from the increasing need for efficient, continuous, and real-time healthcare solutions in today's fast-paced world. With the growing population and rise in lifestyle-related diseases such as heart problems, respiratory disorders, and chronic illnesses, there is a significant demand for systems that can monitor patients' health parameters continuously without requiring constant physical supervision.

Traditional healthcare systems primarily depend on periodic check-ups and manual monitoring, which may not always provide timely insights into a patient's condition. In many critical situations, delays in detecting abnormalities such as irregular heart rate or low oxygen levels can lead to severe health complications. This highlights the necessity of a system that can provide real-time monitoring and instant access to vital health data.

Another major motivating factor is the lack of accessible healthcare facilities in rural and remote areas. Many patients do not have immediate access to hospitals or medical professionals, which increases the risk of delayed diagnosis and treatment. By leveraging IoT technology, it becomes possible to bridge this gap by enabling remote monitoring, where patient data can be accessed anytime and from anywhere through internet-connected devices.

Furthermore, advancements in embedded systems and wireless communication technologies have made it feasible to design compact, low-cost, and energy-efficient healthcare devices. This project aims to utilize these technologies to create a user-friendly system that can be easily deployed for personal as well as clinical use.

The motivation also extends towards reducing the workload of healthcare professionals by automating the monitoring process. By providing continuous and accurate data, the system allows doctors to focus more on diagnosis and treatment rather than routine monitoring tasks.

Overall, this project is driven by the vision of improving healthcare accessibility, enhancing patient safety, and promoting the use of smart technologies in medical applications. It demonstrates how IoT can play a vital role in transforming traditional healthcare systems into more efficient, reliable, and responsive solutions.

1.2 Problem Definition

Traditional healthcare systems rely on manual and periodic monitoring, which is inefficient and time-consuming.

Continuous monitoring of vital parameters like heart rate, SpO_2, and temperature is not always possible in conventional systems.

Delay in detecting abnormal health conditions can lead to serious complications or emergencies.

Patients in rural and remote areas lack access to proper healthcare facilities and real-time monitoring.

Existing health monitoring systems are often expensive, complex, and not portable, making them unsuitable for regular personal use.

There is no efficient system to provide real-time health data remotely to doctors or caregivers.

Lack of automation increases the workload on healthcare professionals.

There is a need for a cost-effective, user-friendly, and reliable system for continuous health monitoring.

The system should enable real-time data transmission using IoT for remote access and monitoring.

1.3 Project Objectives

To design and develop an IoT-based remote patient health monitoring system.

To measure vital health parameters such as heart rate, blood oxygen level (SpO_2), and body temperature using sensors.

To interface sensors with the ESP32 microcontroller for data acquisition and processing.

To display real-time health data locally on an LCD display.

To transmit the collected data over Wi-Fi to a cloud platform (Blynk).

To enable remote monitoring of patient health through a mobile or web dashboard.

To develop a cost-effective and portable system suitable for personal and clinical use.

To ensure continuous and real-time monitoring of patient health parameters.

To reduce the need for manual monitoring and assist healthcare professionals.

To improve patient safety and early detection of abnormalities.

Literature Review

2.1 Pre-requisite Information

The development of an IoT-based health monitoring system requires an understanding of several fundamental technologies and components. The Internet of Things (IoT) plays a crucial role by enabling communication between devices over the internet, allowing real-time data collection and remote monitoring.

Embedded systems form the core of such applications, where microcontrollers like ESP32 are used to interface with sensors and process data. The ESP32 is widely used due to its built-in Wi-Fi capability, low power consumption, and efficient processing performance.

Sensors are essential components in healthcare monitoring systems. The MAX30102 sensor is used for measuring heart rate and blood oxygen levels (SpO_2) using optical sensing techniques, while the DHT11 sensor is used for measuring body temperature. These sensors provide real-time physiological data that can be processed and transmitted.

Wireless communication is another key prerequisite, enabling data transmission from the device to cloud platforms. Platforms like Blynk provide an interface to visualize data on mobile and web dashboards, making remote monitoring feasible.

Additionally, knowledge of programming (Arduino IDE), data acquisition, signal processing, and basic electronics is necessary to integrate hardware and software components effectively.

2.2 Understanding IoT-based remote health monitoring System

The proposed system is an IoT-based remote health monitoring solution designed to measure and transmit vital health parameters in real time. The system consists of sensors, a microcontroller, a display unit, and a cloud-based monitoring platform.

The MAX30102 sensor continuously measures the patient's heart rate and SpO_2 levels, while the DHT11 sensor records body temperature. These sensors are interfaced with the ESP32 microcontroller, which acts as the central processing unit. The ESP32 collects the sensor data, processes it, and prepares it for display and transmission.

The processed data is first displayed locally on an LCD screen, allowing users to view real-time readings directly. Simultaneously, the ESP32 uses its built-in Wi-Fi module to send the data to the Blynk cloud platform.

Through the Blynk application, users and healthcare providers can monitor the patient's health remotely via a mobile or web dashboard. This enables continuous monitoring without requiring the patient to be physically present in a healthcare facility.

The system operates in a continuous loop, where sensor data is repeatedly collected, processed, displayed, and transmitted. This ensures real-time updates and timely detection of any abnormal health conditions.

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2.3 Web Development Technologies

Blynk Cloud Platform:-
Used as the cloud-based backend to store and manage real-time sensor data.

Blynk Web Dashboard:-
Provides a graphical interface to visualize health parameters like heart rate, SpO_2, and temperature in real time.

Mobile Application (Blynk App):-
Enables remote monitoring of patient data through smartphones.

Wi-Fi Communication (ESP32):-
Used for transmitting sensor data from the ESP32 microcontroller to the cloud server over the internet.

HTTP/Cloud Communication Protocols:-
Facilitates data exchange between ESP32 and Blynk cloud servers.

Virtual Pins (Blynk):-
Used to send and receive data between hardware (ESP32) and the web/mobile dashboard.

Embedded C / Arduino Programming:-
Used to write the logic for data acquisition, processing, and communication.

Real-time Data Visualization:-
Implemented using Blynk widgets such as gauges, labels, and charts.

These technologies collectively enable seamless communication between hardware and cloud, allowing real-time remote monitoring of patient health through web and mobile interfaces.

2.4 User Roles and System Workflow

Patient/User role

- Uses the system to monitor personal health parameters
- Views real-time data on LCD and mobile dashboard

- Doctor/Healthcare Provider
- Monitors patient data remotely via web/mobile dashboard
- Analyzes health parameters for diagnosis
- System Administrator (optional)
- Maintains system setup and connectivity
- Ensures proper functioning of hardware and software

System Workflow

- System initializes ESP32, sensors, and LCD
- ESP32 connects to Wi-Fi and Blynk cloud
- Sensors collect data (Heart Rate, SpO_2, Temperature)
- Data is processed by ESP32
- Values are displayed on LCD screen
- Data is transmitted to Blynk cloud
- User/doctor views data on mobile/web dashboard
- Process repeats continuously for real-time monitoring

2.5 Existing Solutions and Gaps

Existing Solutions

- Traditional hospital systems use ECG machines and bedside monitors for patient monitoring.
- Wearable devices such as smartwatches and fitness bands provide basic health tracking.
- Some systems use Bluetooth-based devices for short-range monitoring.
- IoT-based systems using Arduino/Raspberry Pi with cloud platforms are also available.
- Mobile health applications track parameters like heart rate and physical activity.
- Advanced hospital systems provide continuous monitoring with alert mechanisms.

Research Gaps

- Most existing systems are expensive and not affordable for common users.
- Lack of continuous real-time monitoring in low-cost solutions.
- Consumer wearable devices often have limited accuracy and functionality.
- Traditional systems depend on manual observation, leading to delays.
- Limited cloud integration for remote monitoring in many systems.
- Systems are often complex and not user-friendly.
- Poor accessibility in rural and remote areas.
- Most systems monitor limited health parameters, not all vital signs together.
- Lack of portable, compact, and low-cost solutions for personal use.

2.6 Skills and Tools Required

Skills:

- Basic knowledge of Embedded Systems
- Understanding of Internet of Things (IoT) concepts
- Programming skills in Embedded C / Arduino
- Knowledge of sensor interfacing (MAX30102, DHT11)
- Basic understanding of electronics and circuit connections
- Familiarity with Wi-Fi communication and networking
- Ability to work with cloud platforms (Blynk)
- Basic troubleshooting and debugging skills

Tools:

1) Hardware Tools:

- ESP32 Microcontroller
- MAX30102 Sensor (Heart Rate & SpO_2)
- DHT11 Sensor (Temperature)
- 16x2 LCD Display (I2C)
- Breadboard, Jumper Wires, Power Supply

2) Software Tools:

- Arduino IDE (for programming ESP32)
- Blynk Cloud Platform (for IoT dashboard)
- Blynk Mobile App (for remote monitoring)

3) Communication Tools:

- Wi-Fi (for data transmission)
- USB Cable (for programming ESP32)

Background

3.1 Frontend Development

Information:

Frontend development refers to the user interface through which users interact with the system. In this project, the frontend is implemented using the Blynk Web Dashboard and Mobile Application.

The Blynk platform provides various widgets such as gauges, labels, and charts to display real-time health parameters like heart rate, SpO_2, and temperature.

Usage in Project:

- Displays real-time sensor data to users
- Allows remote monitoring via mobile or web
- Provides a user-friendly interface for easy understanding
- Helps doctors/patients to visually analyze health parameters
- Enables continuous tracking without technical knowledge

3.2 Backend Development

Information:

Backend development involves data processing, communication, and cloud integration. In this system, the ESP32 microcontroller and Blynk Cloud act as the backend.

The ESP32 collects data from sensors (MAX30102 and DHT11), processes it, and sends it to the Blynk cloud server via Wi-Fi using virtual pins.

Usage in Project:

- Collects data from health sensors (HR, SpO_2, Temp)
- Processes and converts raw sensor data into usable values
- Sends data to Blynk cloud for storage and visualization

- Maintains real-time communication between hardware and dashboard
- Ensures continuous data flow and system reliability

3.3 Database

Information:

In an IoT-based health monitoring system, a database plays an important role in storing, managing, and retrieving patient health data. In this project, the Blynk Cloud Platform acts as a cloud-based database that stores real-time sensor data.

The database is responsible for handling data received from the ESP32 microcontroller and making it available for visualization on the web and mobile dashboard. Instead of using a traditional database like MySQL, this system uses a cloud-integrated database managed by Blynk, which simplifies data handling and reduces system complexity.

Usage in Project:

- Stores real-time data of heart rate, SpO_2, and temperature
- Maintains continuous data updates from ESP32 via Wi-Fi
- Enables remote access to patient data through web/mobile dashboard
- Uses virtual pins to organize and manage data efficiently
- Supports real-time data visualization using widgets (gauges, charts)
- Eliminates the need for manual database setup and maintenance
- Ensures fast and reliable data communication

System Requirements

4.1 Software Requirements

Arduino IDE

- Used for writing, compiling, and uploading code to the ESP32 microcontroller. It supports Embedded C programming and provides libraries for sensor integration.

Blynk Cloud Platform

- Acts as the IoT cloud server for storing and managing real-time patient data such as heart rate, SpO_2, and temperature.
- Blynk Mobile Application / Web Dashboard
- Provides a user interface to monitor patient health data remotely in real time using smartphones or web browsers.

ESP32 Board Support Package (BSP)

- Required within Arduino IDE to program and configure the ESP32 microcontroller.
- Sensor Libraries
- MAX30102 library for heart rate and SpO_2 measurement
- DHT library for temperature sensing
- These libraries simplify sensor data acquisition and processing.

4.2 Hardware Requirements

The implementation of the IoT-Based Remote Patient Health Monitoring System requires the following hardware components for data acquisition, processing, and display:

Hardware Components Used

- ESP32 / NodeMCU Microcontroller

  Acts as the main processing unit of the system. It collects data from sensors, processes it, and transmits it to the cloud via Wi-Fi.

- MAX30102 Sensor

  Used to measure heart rate and blood oxygen level (SpO_2) using optical sensing technology.

- DHT11 Sensor

  Used to measure body temperature of the patient.

- 16x2 LCD Display (I2C)

  Displays real-time health parameters such as heart rate, SpO_2, and temperature locally.

- Breadboard

  Used for assembling the circuit without soldering.

- Jumper Wires

  Used for connecting different components in the circuit.

- Power Supply / USB Cable

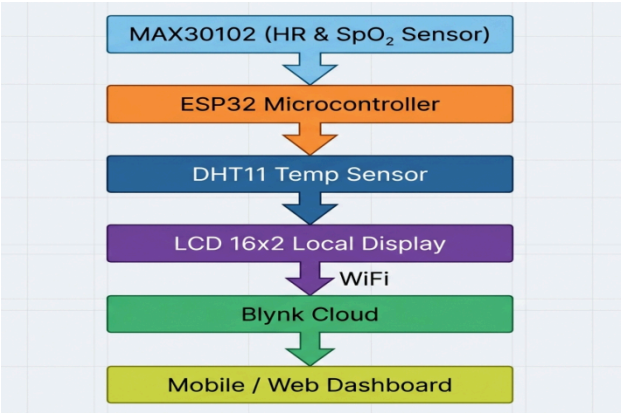
  Provides power to the ESP32 and connected components.

System Design

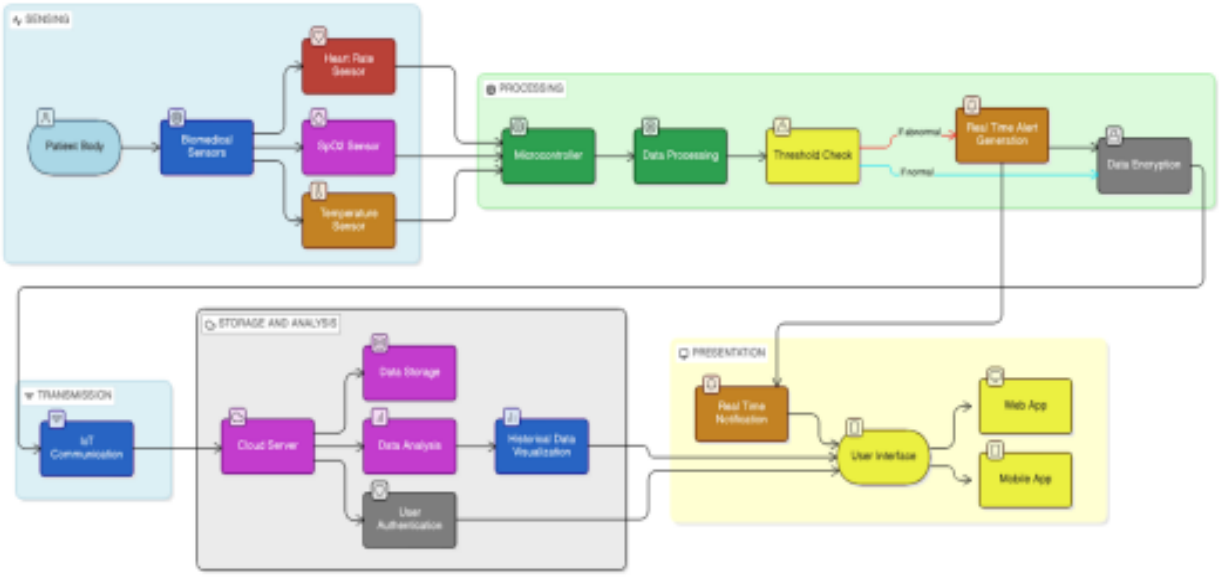
List of Figures

Figure No.	Title
Figure 1.1	Overall System Architecture of the IoT Based Remote Patient Health Monitoring System
Figure 1.2	Data Flow for Monitoring Patients Health
Figure 1.2	System Working

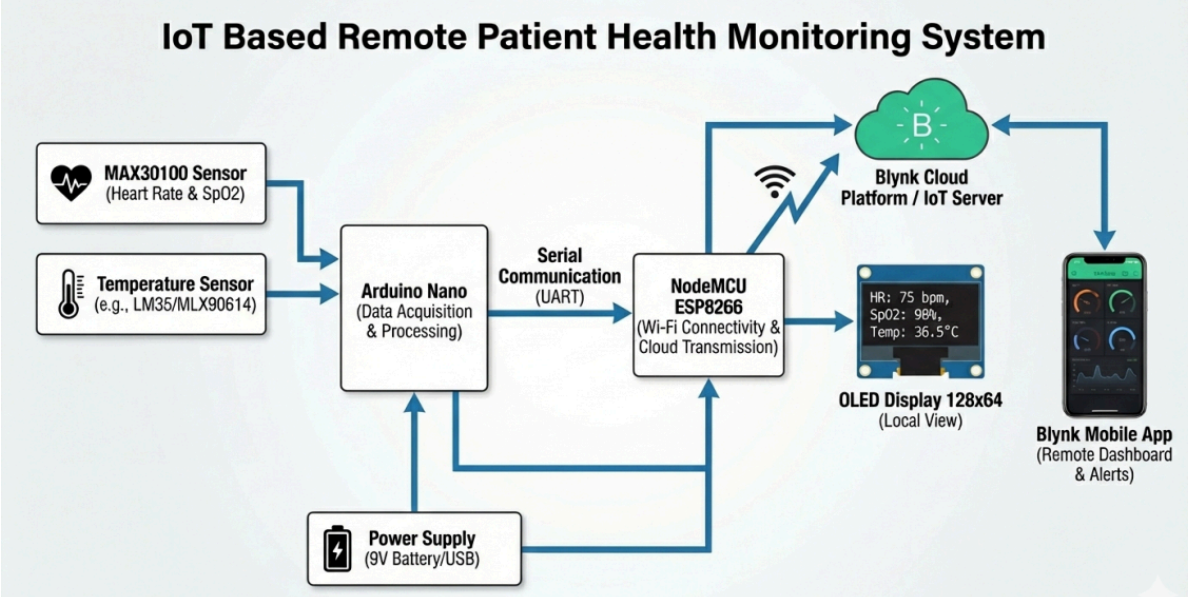
5.1 System Architecture of the IoT Based Remote Patient Health Monitoring System



5.2 Dataflow in system



5.3 System Working



System Implementation

Our system uses IoT, embedded systems, sensors, and cloud platforms to monitor patient health in real time and provide remote access through mobile and web applications.

1. Overview

The proposed system, IoT-Based Remote Patient Health Monitoring System, is designed to monitor vital health parameters such as heart rate, blood oxygen level (SpO_2), and body temperature in real time. The system integrates sensors, a microcontroller, and cloud technology to provide continuous and remote health monitoring.

In this system, sensors like MAX30102 and DHT11 are used to collect physiological data from the patient. The collected data is processed by the ESP32 microcontroller, which acts as the central unit. The processed data is displayed locally on a 16x2 LCD screen and simultaneously transmitted to a cloud platform using Wi-Fi.

The cloud platform (Blynk/ThingSpeak) stores and visualizes the data, allowing users and healthcare professionals to monitor patient health remotely through a mobile or web

dashboard. The system operates continuously, ensuring real-time updates and enabling early detection of abnormal conditions.

2. Technologies Used

- Internet of Things (IoT)

  Enables communication between hardware devices and cloud platforms for remote monitoring.

- Embedded Systems

  ESP32 microcontroller is used for processing sensor data and controlling system operations.

- Sensor Technology
- MAX30102 → Heart Rate & SpO_2 measurement
- DHT11 → Temperature measurement
- Wireless Communication (Wi-Fi)

  Used to transmit data from ESP32 to cloud platforms in real time.

- Cloud Computing (Blynk / ThingSpeak)

  Used for data storage, processing, and visualization through dashboards.

- Mobile & Web Applications

  Blynk app/web dashboard used for remote monitoring of patient data.

- Programming (Embedded C / Arduino)

  Used to implement system logic, data processing, and communication.

3. Implementation Summary

The IoT-Based Remote Patient Health Monitoring System is implemented using sensors, a microcontroller, and cloud technology to provide continuous and real-time monitoring of patient health parameters.

The system uses an ESP32/NodeMCU microcontroller to collect data from sensors such as the MAX30102 (for heart rate and SpO_2) and a temperature sensor. The collected data is processed and filtered by the microcontroller to ensure accurate readings. The system then connects to the internet using Wi-Fi for data transmission.

The measured health parameters are analyzed and compared with normal threshold values. If any abnormal condition is detected, alerts can be generated to notify users. The processed data is transmitted to cloud platforms like Blynk or ThingSpeak, where it is stored and displayed on a mobile or web dashboard for remote monitoring.

In addition to cloud monitoring, the system also provides local display of real-time data on an LCD, allowing immediate access to readings. The entire system operates continuously in a loop, ensuring real-time updates and improved patient safety.

4. Deployment

The deployment of the IoT-Based Remote Patient Health Monitoring System involves setting up hardware, uploading code, and configuring cloud connectivity.The hardware components such as ESP32/NodeMCU, MAX30102 sensor, temperature sensor, and LCD are connected using a breadboard. The system is powered via USB.After setup, the ESP32 connects to a Wi-Fi network and communicates with cloud platforms like Blynk or ThingSpeak. The dashboard is configured to display real-time health data using widgets.Once deployed, the system continuously collects, processes, and sends data to the cloud, allowing users to monitor patient health remotely through a mobile or web interface.

System Testing

1. Objective

The objective of system testing is to verify that the IoT-Based Remote Patient Health Monitoring System functions correctly and meets all specified requirements. It ensures that all hardware and software components work together efficiently as an integrated system.

2. Types of Testing Performed

- Unit Testing
  Individual components like MAX30102 sensor, DHT11 sensor, and LCD were tested separately.

- Integration Testing
  Verified proper interaction between sensors, ESP32 microcontroller, LCD, and cloud platform.

- System Testing
  Tested the complete system to ensure all modules work together correctly.

- Functional Testing
  Checked whether the system correctly measures heart rate, SpO_2, and temperature.

- Connectivity Testing
  Ensured stable Wi-Fi connection and proper data transmission to Blynk/ThingSpeak.

- Performance Testing
  Evaluated system response time and real-time data updating.

- Accuracy Testing
  Verified sensor readings for correctness and consistency.

- User Interface Testing

Checked proper display of data on LCD and mobile/web dashboard.

- Stress Testing
  Tested system under continuous operation for stability.

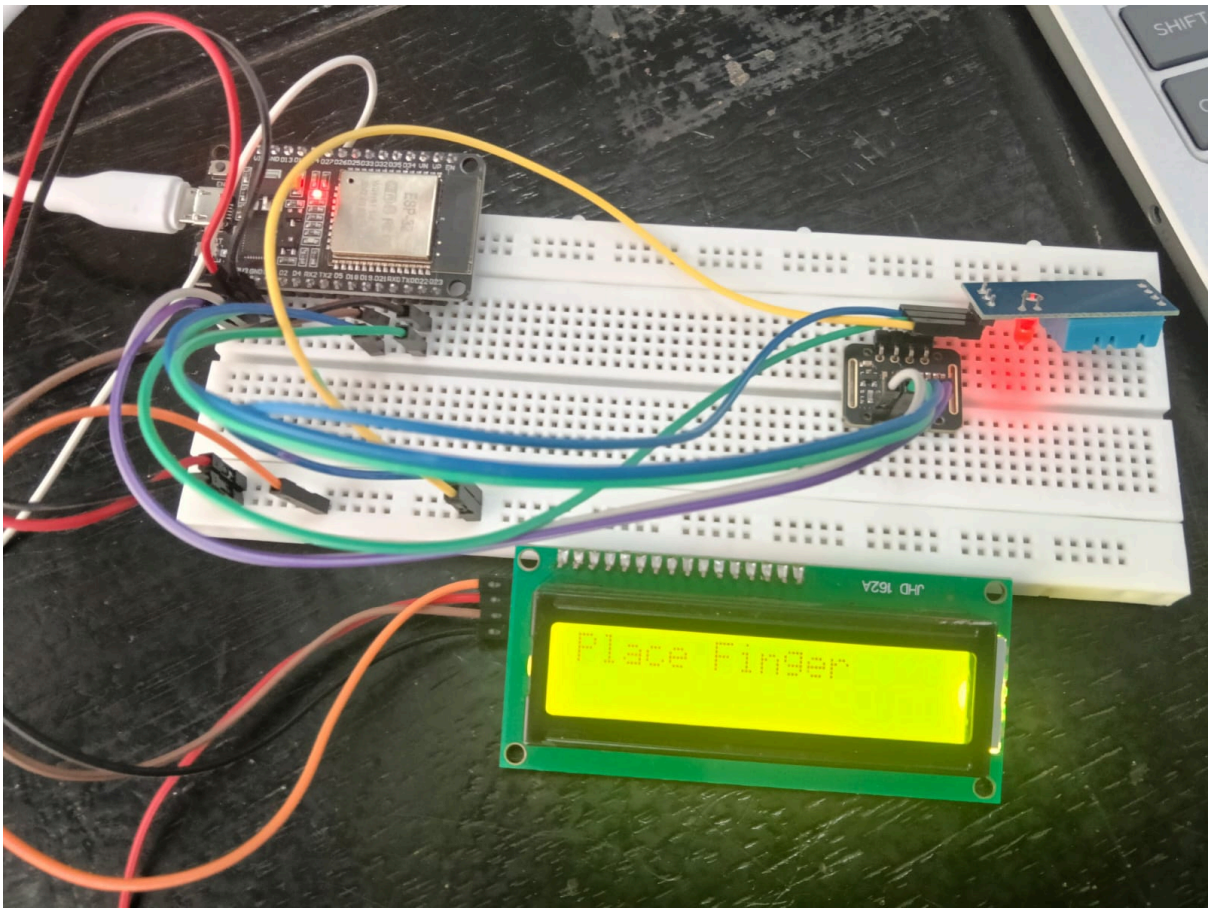
3. Test Scenarios

The system was tested under various scenarios including initialization of the ESP32 and sensors, detection of finger placement on the MAX30102 sensor, and accurate measurement of heart rate, SpO_2, and temperature. The display of real-time data on the LCD and proper data transmission to the Blynk/ThingSpeak cloud were also verified. Additionally, Wi-Fi connectivity, dashboard updates, alert generation for abnormal values, and continuous system performance were tested to ensure reliability and accuracy.

4. Test Results

The system was successfully tested under various conditions to evaluate its performance and reliability. The sensors were able to accurately measure heart rate, SpO_2, and temperature values. The ESP32 microcontroller processed the data efficiently and transmitted it to the cloud platform without delays.

The LCD displayed real-time readings correctly, and the Blynk/ThingSpeak dashboard showed continuous updates of health parameters. Wi-Fi connectivity was stable, ensuring proper data transmission. The system was also able to detect abnormal values and respond accordingly. Overall, the system performed reliably during continuous operation.



5. Conclusion

The IoT-Based Remote Patient Health Monitoring System was successfully developed to monitor vital parameters in real time. The system is reliable, cost-effective, and enables remote access to patient data, improving healthcare monitoring and early detection of health issues.

Experimental Results

The proposed IoT-Based Remote Patient Health Monitoring System was experimentally tested under various conditions to evaluate its performance, accuracy, and reliability. The system was able to successfully measure and monitor vital health parameters such as heart rate, blood oxygen saturation (SpO_2), and body temperature in real time.

During testing, the MAX30102 sensor provided stable and consistent readings for heart rate and SpO_2 when the finger was properly placed. Minor fluctuations were observed due to motion and improper finger placement, but these were effectively handled through data filtering and validation in the system. The temperature sensor (DHT11) also produced acceptable readings within its operational accuracy range.

The ESP32 microcontroller efficiently processed sensor data and maintained a stable Wi-Fi connection for continuous data transmission. The system demonstrated reliable performance in sending real-time data to the Blynk/ThingSpeak cloud platform, where values were accurately displayed on the mobile and web dashboard without significant delay.

The LCD display successfully showed real-time readings, allowing immediate local monitoring. The system also responded appropriately to abnormal conditions by identifying out-of-range values, demonstrating its capability for basic alert functionality.

Continuous operation testing showed that the system could run for extended periods without failure, indicating good stability. Overall, the experimental results confirm that the system is capable of providing accurate, reliable, and real-time remote health monitoring, making it suitable for practical healthcare applications.

Conclusion

The IoT-Based Remote Patient Health Monitoring System has been successfully designed, developed, and implemented to provide continuous and real-time monitoring of vital health parameters such as heart rate, blood oxygen saturation (SpO_2), and body temperature. The

system effectively integrates biomedical sensors, a microcontroller (ESP32), and cloud-based IoT platforms to create a reliable and efficient healthcare monitoring solution.

The experimental results demonstrate that the system is capable of accurately acquiring and processing physiological data, and transmitting it to cloud platforms such as Blynk or ThingSpeak with minimal delay. The real-time visualization of data on mobile and web dashboards enables remote monitoring, making it highly useful for patients, caregivers, and healthcare professionals.

The system also provides local feedback through an LCD display, ensuring accessibility even without internet connectivity. Its ability to continuously monitor data and detect abnormal conditions enhances patient safety and allows for timely medical intervention.

Moreover, the proposed system is cost-effective, portable, and easy to use, making it suitable for both home-based healthcare and small clinical setups. It reduces the dependency on manual monitoring and helps in improving the overall efficiency of healthcare services.

In conclusion, this project successfully demonstrates the practical application of IoT in healthcare, offering a scalable and efficient solution for remote patient monitoring and contributing towards the advancement of smart healthcare systems.

Future Scope

The IoT-Based Remote Patient Health Monitoring System can be further improved and expanded with advanced features and technologies to enhance its effectiveness and usability.

Future Enhancements

- Integration of additional sensors such as ECG, blood pressure, and glucose monitoring
- Implementation of AI/ML algorithms for early disease prediction
- Development of a dedicated mobile application with better UI/UX
- Addition of real-time alerts via SMS, email, or app notifications
- Enhancement of data security using encryption techniques
- Integration with hospital management systems and EHR
- Development of wearable devices for continuous monitoring
- Use of cloud analytics for tracking long-term health trends
- Support for multi-patient monitoring systems
- Integration with telemedicine platforms for remote consultation

Contribution

Sr. No.	Member Name	Technical Contribution	Non-Technical Contribution
1	Vedant Awachar	Developed core system using ESP32; implemented sensor integration (MAX30102, DHT11); handled IoT communication with Blynk/Thing Speak; Optimized data processing and real-time transmission; managed system integration and debugging	Prepared major portions of project report; designed system architecture and flow diagrams; contributed to presentation and documentation
2	Yashraj Kadam	Worked on hardware setup and circuit connections; implemented LCD display and real-time data visualization; assisted in coding and Wi-Fi connectivity; performed system testing and validation	Prepared documentation sections (testing, results, conclusion); created diagrams and report formatting; contributed to presentation slides
3	Sarvesh Botalji	Assisted in sensor interfacing and data collection; supported cloud integration and dashboard setup; helped in testing system performance	Contributed to literature review; assisted in report writing and data collection; helped in presentation preparation
4	Anant Darade	Assisted in hardware assembly and troubleshooting; supported code implementation and system testing	Helped in documentation, formatting, and compilation of report; contributed to research and content organization

List of Abbreviations

- IoT – Internet of Things
- ESP32 – Espressif 32-bit Microcontroller
- SpO_2 – Peripheral Capillary Oxygen Saturation
- HR – Heart Rate
- BPM – Beats Per Minute
- LCD – Liquid Crystal Display
- I2C – Inter-Integrated Circuit
- DHT – Digital Humidity and Temperature Sensor
- Wi-Fi – Wireless Fidelity
- API – Application Programming Interface
- UI – User Interface
- UX – User Experience
- EHR – Electronic Health Records
- ECG – Electrocardiogram
- ML – Machine Learning
- AI – Artificial Intelligence
- USB – Universal Serial Bus
- IDE – Integrated Development Environment
- TCP/IP – Transmission Control Protocol / Internet Protocol

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